

## 15.2 Open-Circuit Time Constants as a Design Aid

Although the computer solution discussed above provides a simple method for analyzing multistage circuits, it unfortunately provides almost no insight into *design*. For example, the computer solution gives no guidance about how best to redesign the cascode circuit to increase its bandwidth, because the analysis does not identify the circuit elements primarily responsible for the bandwidth limitation. Thus we must develop additional analysis methods that give such insight. However, because we have computer methods to provide accurate answers, the additional methods can be quite approximate, hence simple and rapid.

One such method involves calculating  $\omega_h$  from a group of easily computed network time constants. Unfortunately, it takes a good deal longer to derive on a rational basis the relation between  $\omega_h$  and these time constants than it does to make the actual calculation. The proof is in two parts. The first part establishes the relation between  $\omega_h$  and the first two terms of the transfer function denominator polynomial, and the second relates the same denominator terms to the time constants.\*

### 15.2.1 Relationship between $\omega_h$ and the First Two Denominator Terms of the Transfer Function

If we multiply out the denominator of Eq. 15.5, and again assume (as in Section 15.1.3) that the zeros in that equation will not influence the bandwidth calculation, we obtain an expression of the form

$$\frac{V_o}{V_i} = \frac{K'}{a_0 + a_1s + a_2s^2 + a_3s^3 + s^4} \quad (15.20)$$

The magnitude of this expression at  $s = j\omega_h$  is

$$\left| \frac{V_o}{V_i} \right|_{s=j\omega} = \left| \frac{K'}{a_0 + ja_1\omega_h - a_2\omega_h^2 - ja_3\omega_h^3 + \omega_h^4} \right|$$

Again assuming that the network natural frequencies  $s_d$ ,  $s_e$ ,  $s_f$ , and  $s_g$  are real, we can apply the same argument as that following Eq. 15.12 to find an approximate value for  $\omega_h$ . That is, we square both sides of the equation, and solve for  $\omega_h$  by noting from Eq. 15.20 that at the 0.707 frequency  $\omega_h$ ,

$$\left| \frac{V_o}{V_i} \right|_{s=j\omega_h}^2 = \frac{(K')^2}{2a_0^2}$$

If we discard all terms above  $\omega_h^2$ , we obtain

$$\omega_h \simeq \frac{a_0}{\sqrt{a_1^2 - 2a_0a_2}} \quad (15.21)$$

\* In an introductory course, the proofs in the next two subsections can be omitted without loss in continuity, because the important results of the proofs have been restated at the beginning of Section 15.2.3.

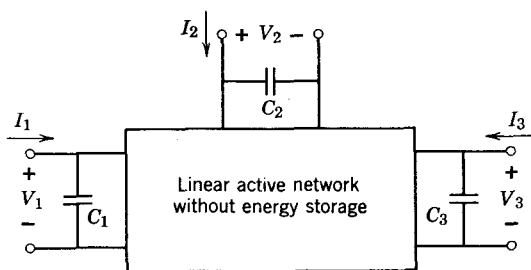
(see Problem P15.2). A simpler but less accurate approximation results if we neglect the  $a_0 a_2$  term in Eq. 15.21:

$$\omega_h \simeq \frac{a_0}{a_1} \quad (15.22)$$

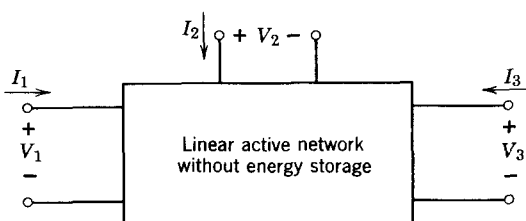
This approximation is particularly useful because it gives  $\omega_h$  in terms of the ratio of the first two coefficients in Eq. 15.20, and this ratio can usually be found by inspection of the network, as we shall now show.

### 15.2.2 Finding $a_1/a_0$ and $a_{n-1}/a_n$ by Inspection of the Network

To find the general relationship between  $a_1/a_0$  and the network elements, we consider a linear active network that contains  $n$  capacitors and no other energy storage. For simplicity, we develop the relation for  $a_1/a_0$  for the three-capacitor network of Fig. 15.3a, recognizing that the proof can be readily generalized. Disregarding any terminal pairs originally associated with the network, we assign a voltage and current to each of the terminal pairs associated with the capacitors, as shown in the figure.



(a) Capacitors identified by terminal pair



(b) Network with capacitors removed

**Figure 15.3** Representation of a linear active network containing three capacitors.

We subsequently have need for two sets of low-frequency parameters:

1. The short-circuit conductances, that is, the conductance measured at each port with all other ports short-circuited.
2. The open-circuit resistances, that is, the resistance measured at each port with all other ports open-circuited.

To find these, we remove the capacitors, as shown in Fig. 15.3b, and represent the linear active network that remains by the node equations given in Eqs. 15.23. Any *independent* sources within the box are assumed to be set to zero; the effects of any *dependent* sources within the box must be included in the equations:

$$I_1 = g_{11}V_1 + g_{12}V_2 + g_{13}V_3 \quad (15.23a)$$

$$I_2 = g_{21}V_1 + g_{22}V_2 + g_{23}V_3 \quad (15.23b)$$

$$I_3 = g_{31}V_1 + g_{32}V_2 + g_{33}V_3 \quad (15.23c)$$

We can find the short-circuit conductances  $G_{js}$  by noting that Eqs. 15.23 are in  $y$ -parameter form (see Section 12.1.1). Thus from the definition of  $y_i$  or  $y_o$  (Eqs. 12.2a and b), the short-circuit conductance at terminal-pair  $j$  is just  $g_{jj}$ :

$$G_{js} = g_{jj} \quad (15.24)$$

To find the open-circuit resistances, designated by  $R_{jo}$ , we solve Eqs. 15.23 for  $V_j$  in terms of  $I_j$ , with all other currents equal to zero. For convenience we use the shorthand notation of determinants. We designate the  $g$  determinant as

$$\Delta_g = \begin{vmatrix} g_{11} & g_{12} & g_{13} \\ g_{21} & g_{22} & g_{23} \\ g_{31} & g_{32} & g_{33} \end{vmatrix} \quad (15.25)$$

and the cofactors by  $(\Delta_g)_{jk}$ . For example,

$$(\Delta_g)_{11} = \begin{vmatrix} g_{22} & g_{23} \\ g_{32} & g_{33} \end{vmatrix} \quad (15.26)$$

With this notation Cramer's rule yields

$$V_j = I_j \frac{(\Delta_g)_{jj}}{\Delta_g}$$

Thus the open-circuit resistance at terminal-pair  $j$  is

$$R_{jo} = \frac{(\Delta_g)_{jj}}{\Delta_g} \quad (15.27)$$

When the capacitors are present at the terminal pairs, as shown in Fig. 15.3a, the complete network can be represented by a set of node equations with the following admittance determinant, which is obtained by adding the appropriate capacitive susceptances to the conductances on the principal diagonal of  $\Delta_g$ :

$$\Delta_y = \begin{vmatrix} g_{11} + sC_1 & g_{12} & g_{13} \\ g_{21} & g_{22} + sC_2 & g_{23} \\ g_{31} & g_{32} & g_{33} + sC_3 \end{vmatrix} \quad (15.28)$$

The natural frequencies of the complete network are those frequencies at which voltages can exist at the terminals when  $I_1 = I_2 = I_3 = 0$ . This condition can be met only if  $\Delta_y = 0$ . Inspection of Eq. 15.28 shows that  $\Delta_y$  will contain, in general, all powers of  $s$  from  $s^0$  up to  $s^3$ . If there are  $n$  capacitors and no capacitor loops, the highest power of  $s$  in the characteristic equation is  $n$ :

$$\Delta_y = 0 = a_0 + a_1s + a_2s^2 + a_3s^3 \quad (15.29)$$

If all natural frequencies have negative real parts, the network is stable, and there then will be no missing coefficients in the middle of the equation. That is, in our example with three capacitors,  $a_3$  can be zero if there is a capacitor loop, but if  $a_3$  and  $a_0$  are nonzero, then  $a_1$  and  $a_2$  must also be nonzero.

Some of the coefficients in Eq. 15.29 can be identified relatively easily by inspection of  $\Delta_y$ , Eq. 15.28. The coefficient  $a_0$ , the constant term, is simply  $\Delta_y$  when  $s = 0$ , or equivalently,  $a_0$  is equal to  $\Delta$  for the network with all the capacitors removed. That is,

$$a_0 = \Delta_g \quad (15.30)$$

The coefficient  $a_1$  (that of the first power of  $s$ ) must be the sum of three terms, each of which contains one capacitance multiplied by the cofactor of the corresponding element in the conductance determinant. On this basis,  $a_1$  is

$$a_1 = C_1(\Delta_g)_{11} + C_2(\Delta_g)_{22} + C_3(\Delta_g)_{33} \quad (15.31)$$

Thus the desired ratio  $a_1/a_0$  is

$$\frac{a_1}{a_0} = C_1 \frac{(\Delta_g)_{11}}{\Delta_g} + C_2 \frac{(\Delta_g)_{22}}{\Delta_g} + C_3 \frac{(\Delta_g)_{33}}{\Delta_g} \quad (15.32)$$

However,  $(\Delta_g)_{jj}/\Delta_g$  is the open-circuit resistance seen by  $C_j$ , as shown in Eq. 15.27. Consequently,

$$\frac{a_1}{a_0} = R_{1o}C_1 + R_{2o}C_2 + R_{3o}C_3 = \sum_j \tau_{jo} \quad (15.33)$$

where  $\tau_{jo}$  denotes the time constant of the  $j$ th capacitor calculated with all other capacitors *open-circuited*. Open-circuiting all capacitors but one usually breaks a complicated network into several simpler subnetworks. Hence the open-circuit resistances  $R_{jo}$  can often be found by inspection (*no* energy-storage elements are involved in this calculation).

We also derive at this point a second relation among the coefficients of the denominator polynomial, which is useful for finding the low-frequency 0.707 point of an amplifier. In Eq. 15.29, the coefficient  $a_3$  (or in general the coefficient of the  $s^n$  term) results from multiplying the susceptible parts of the elements on the principal diagonal:

$$a_3 = C_1C_2C_3 \quad (15.34)$$

The coefficient  $a_2$  (in general the coefficient of the  $s^{n-1}$  term), which also results from multiplying terms on the principal diagonal, contains three terms, each of which involves all capacitors except one:

$$a_2 = C_2 C_3 g_{11} + C_1 C_3 g_{22} + C_1 C_2 g_{33} \quad (15.35)$$

Hence

$$\frac{a_2}{a_3} = \frac{g_{11}}{C_1} + \frac{g_{22}}{C_2} + \frac{g_{33}}{C_3} \quad (15.36)$$

$$\equiv \sum_j \frac{1}{\tau_{js}} \quad (15.37)$$

where  $\tau_{js}$  denotes the time constant of the  $j$ th capacitor calculated with all other capacitors *shorted* (see Eq. 15.24). Shorting all capacitors but one usually collapses a complicated network sufficiently so that again the time constants can be determined by inspection.

### 15.2.3 Relation between $\omega_h$ and the Open-Circuit Time Constants

In the preceding section we proved that for an  $RC$  network with no capacitor loops, the ratio of the coefficient of the  $s$  term to the constant term in the denominator of the gain expression is (in general)

$$\frac{a_1}{a_0} = R_{1o}C_1 + R_{2o}C_2 + \cdots + R_{no}C_n = \sum_j \tau_{jo} \quad (15.38)$$

where  $R_{jo}$  denotes the resistance seen by the  $j$ th capacitor and  $\tau_{jo}$  denotes the time constant of the  $j$ th capacitor, both calculated with all other capacitors open-circuited. In Section 15.2.1 we showed that a rough approximation to the 0.707 frequency  $\omega_h$  could be found from the same two coefficients:

$$\omega_h \simeq \frac{a_0}{a_1} \quad (15.39)$$

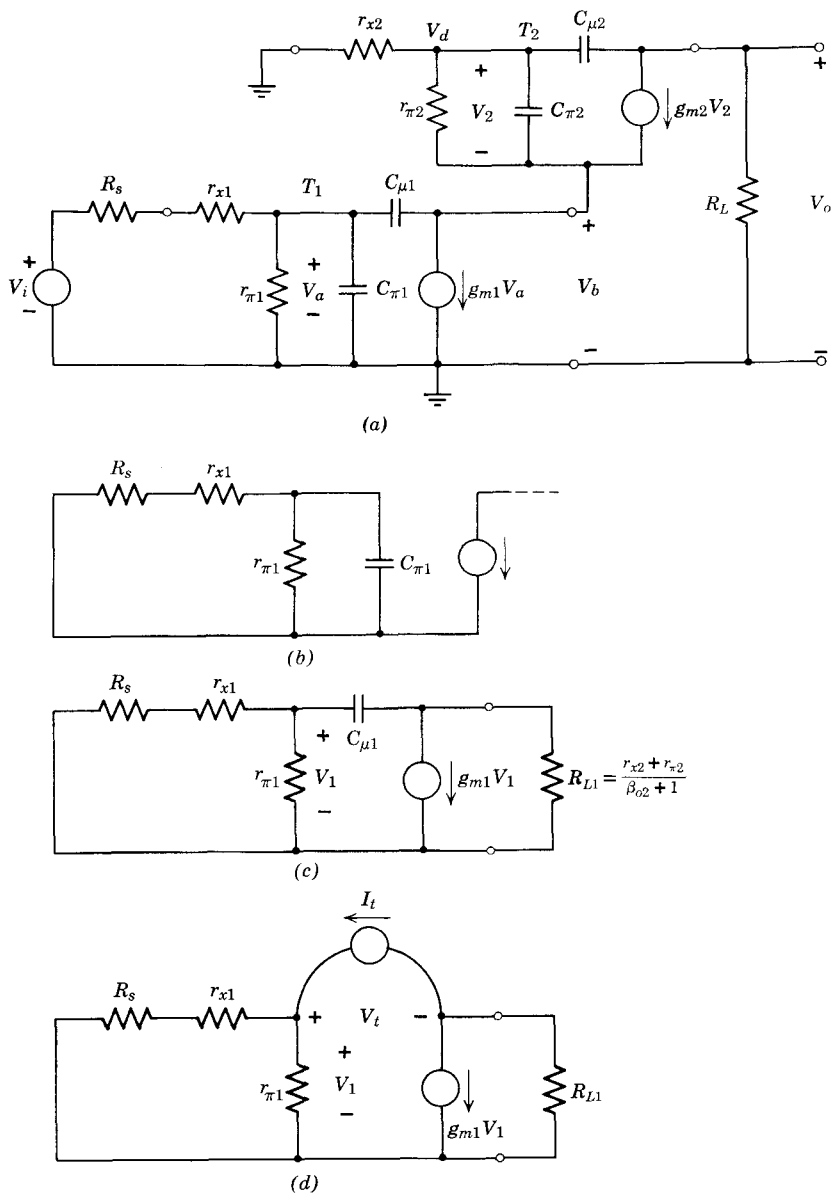
Combining the two results, we obtain a very simple relationship between  $\omega_h$  and the open-circuit time constants which provides valuable design insight:

$$\omega_h \simeq \frac{1}{\sum \tau_{jo}} \quad (15.40)$$

This relation will allow us to localize to some extent the circuit elements limiting the amplifier bandwidth, by indicating which open-circuit time constant is dominant in the bandwidth calculation. Let us illustrate this by calculating these time constants for the cascode circuit introduced in Sections 15.1.1 and 15.1.2.

### 15.2.4 Example

For convenience the incremental circuit for the cascode amplifier has been repeated in Fig. 15.4a. When all capacitors but  $C_{\pi 1}$  are open-circuited, as



**Figure 15.4** Calculation of open-circuit time constants associated with transistor  $T_1$  in the cascode circuit.

shown in Fig. 15.4b, the input network becomes decoupled from the remainder of the circuit, thereby making it possible to calculate  $\tau_{1o}$  by inspection. In units of kilohms, picofarads, and nanoseconds,

$$\begin{aligned} R_{1o} &= r_{\pi 1} \parallel (r_{x1} + R_s) \\ &= 0.25 \parallel 0.22 = 0.117 \text{ kohm} \end{aligned}$$

Thus

$$\begin{aligned} \tau_{1o} &= C_{\pi 1} R_{1o} \\ &= 100(0.117) = 11.7 \text{ nsec} \end{aligned}$$

Figure 15.4c shows the network that results when only capacitor  $C_{\mu 1}$  remains in the circuit. The input impedance of transistor  $T_2$  reduces to

$$\begin{aligned} R_{L1} &= \frac{r_{x2} + r_{\pi 2}}{\beta_{o2} + 1} \\ &\simeq \frac{0.270}{100} = 0.0027 \text{ kohm} \end{aligned}$$

under these conditions. Even with this simplification, the calculation of  $R_{2o}$ , the resistance facing  $C_{\mu 1}$ , cannot be done by inspection. A fairly straightforward way of finding  $R_{2o}$  is to apply a test current  $I_t$  to the terminals in question, as shown in Fig. 15.4d, and calculate the resultant voltage. We choose a current source for the test source because this will allow us to calculate  $V_1$  most readily. By inspection

$$\begin{aligned} V_1 &= I_t [r_{\pi 1} \parallel (r_{x1} + R_s)] \\ &= I_t R_{1o} \end{aligned}$$

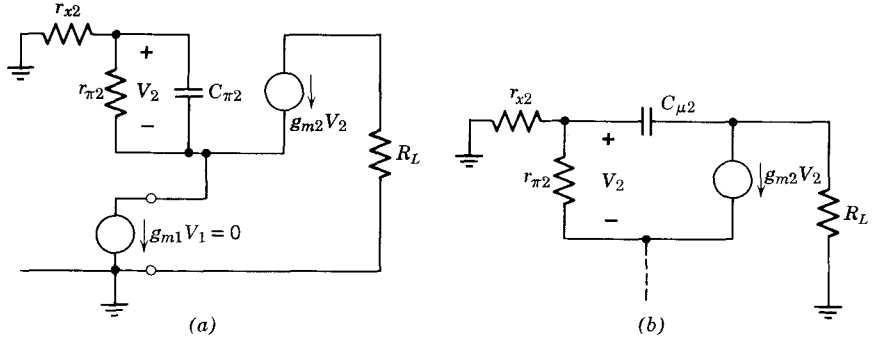
Thus

$$\begin{aligned} \frac{V_t}{I_t} = R_{2o} &= R_{1o} + g_m R_{1o} R_{L1} + R_{L1} \\ &= 0.117 + 400(0.117)0.0027 + 0.0027 \\ &= 0.243 \text{ kohm} \end{aligned}$$

Hence

$$\begin{aligned} \tau_{2o} &= C_{\mu 1} R_{2o} = 5(0.243) \\ &= 1.22 \text{ nsec} \end{aligned}$$

The circuit in Fig. 15.5a results when all capacitors in Fig. 15.4a except  $C_{\pi 2}$  are set to zero. The simplest way to find  $R_{3o}$  facing  $C_{\pi 2}$  is to set  $C_{\pi 2}$  to zero, apply a voltage source  $V_t$  across the capacitor terminals, and calculate the resulting current  $I_t$  (see Problem P15.3). The current generator  $g_{m1} V_1$  is



**Figure 15.5** Calculation of  $R_{3o}$  and  $R_{4o}$  associated with transistor  $T_2$  in the cascode circuit.

not excited under these conditions, hence appears as an open circuit. However, generator  $g_{m2}V_2$  is excited and is in fact the dominant factor in  $G_{3o}$ :

$$\begin{aligned}\frac{I_t}{V_t} &= G_{3o} = g_{\pi 2} + g_{m2} = 4 + 400 \\ &= 404 \text{ mmhos}\end{aligned}$$

That is,

$$R_{3o} = 2.47 \text{ ohms}$$

Hence

$$\begin{aligned}\tau_{3o} &= \frac{C_{\pi 2}}{G_{3o}} = \frac{100}{404} \\ &= 0.247 \text{ nsec}\end{aligned}$$

Figure 15.5b can be used to calculate the fourth open-circuit time constant. Applying an excitation across capacitor  $C_{\mu 2}$  cannot cause any current to flow through the  $r_{\pi 2}, g_{m2}$  branch because with generator  $g_{m1}V_1$  an open circuit, we must have

$$V_2 = -g_{m2}V_2r_{\pi 2}$$

and for positive coefficients the only solution to this equation is  $V_2 = 0$ . Hence

$$\begin{aligned}R_{4o} &= r_{x2} + R_L = 0.02 + 0.2 \\ &= 0.22 \text{ kohm}\end{aligned}$$

and

$$\begin{aligned}\tau_{4o} &= R_{4o}C_{\mu 2} = 0.22(5) \\ &= 1.1 \text{ nsec}\end{aligned}$$



Thus the sum of the open-circuit time constants is

$$\begin{aligned}\Sigma\tau_{jo} &= 11.7 + 1.22 + 0.25 + 1.1 \\ &= 14.27 \text{ nsec}\end{aligned}$$

Now, from Eq. 15.40,

$$\omega_h \simeq \frac{1}{14.27} = 0.07 \text{ Grad/sec}$$

(a result that is about 13% lower than the more accurate calculation in Section 15.1.3).

The important insight that we gain from this calculation is that the dominant factor limiting the bandwidth of this amplifier is the time constant  $\tau_{1o}$ . Hence reducing  $C_{\pi 1}$ ,  $r_{x1}$ ,  $r_{\pi 1}$ , or  $R_s$  will pay big dividends in increased bandwidth, whereas reducing any other resistor or capacitor values will bring about only minimal changes.

To verify this conclusion, two additional computer calculations were made on the amplifier of Fig. 15.4a. In the first,  $C_{\pi 1}$  was reduced from 100 to 50 pf, and in the second,  $C_{\pi 2}$  was reduced from 100 to 50 pf. The results are summarized in Table 15.1, along with the results of the original computer calculation.

**Table 15.1**

|            | $C_{\pi 1} = 100 \text{ pf}$<br>$C_{\pi 2} = 100 \text{ pf}$ | $C_{\pi 1} = 50 \text{ pf}$<br>$C_{\pi 2} = 100 \text{ pf}$ | $C_{\pi 1} = 100 \text{ pf}$<br>$C_{\pi 2} = 50 \text{ pf}$ |
|------------|--------------------------------------------------------------|-------------------------------------------------------------|-------------------------------------------------------------|
| $s_d$      | $-0.0806 \text{ nsec}^{-1}$                                  | $-0.17 \text{ nsec}^{-1}$                                   | $-0.0803$                                                   |
| $s_e$      | $-0.64$                                                      | $-0.44$                                                     | $-0.67$                                                     |
| $s_f$      | $-4.05$                                                      | $-4.45$                                                     | $-7.20$                                                     |
| $s_g$      | $-16.5$                                                      | $-20.8$                                                     | $-17.9$                                                     |
| $\omega_h$ | $79.9 \text{ Mrad/sec}$                                      | $159 \text{ Mrad/sec}$                                      | $79.7 \text{ Mrad/sec}$                                     |

Thus when  $C_{\pi 1}$  is halved, the bandwidth nearly doubles, as predicted. Also as predicted, when  $C_{\pi 2}$  is halved, no significant bandwidth improvement results (although the very slight bandwidth decrease comes as a bit of a surprise).

To further reinforce the concept of open-circuit time constants, the reader is urged to calculate  $\tau_{1o}$  and  $\tau_{2o}$  for the single-stage common-emitter amplifier discussed at the end of Section 15.1.3 and to compare the results with those obtained for the cascode (see Problem P15.4).

### 15.2.5 A Useful Computational Check

It is a simple matter to check for consistency between the open-circuit time constants and the natural frequencies. By equating the ratio  $a_1/a_0$  in Eq.

15.20 to the ratio of the first two terms in the multiplied-out denominator of Eq. 15.11, we find

$$\frac{a_1}{a_0} = - \left[ \frac{1}{s_d} + \frac{1}{s_e} + \frac{1}{s_f} + \frac{1}{s_g} \right] \quad (15.41)$$

Hence, from Eq. 15.38, we find

$$- \left[ \frac{1}{s_d} + \frac{1}{s_e} + \frac{1}{s_f} + \frac{1}{s_g} \right] = \sum_j \tau_{jo} \quad (15.42)$$

or, in general,

$$- \sum_j \frac{1}{s_j} = \sum_j \tau_{jo} \quad (15.43)$$

Note that this is an *exact* expression, hence can be used as a check on the calculations to as many significant figures as one wishes to use in carrying out the calculations (see Problems P15.5, P15.6, and P15.7). In the present example, for  $C_{\pi 1} = C_{\pi 2} = 100$  pf,  $-\Sigma 1/s_j = 14.3$ , which agrees within slide-rule accuracy with  $\Sigma \tau_{jo} = 14.27$  as calculated on page 539.

### 15.2.6 Multistage Design Procedure

We are now in a position to outline a relatively simple method for designing multistage amplifiers to meet some given gain and bandwidth specification. It should be emphasized that the method suggested below is *not* unique. Many different design procedures are possible, and the reader is urged to explore other methods, or, better yet, devise his own.

In the design procedure we shall follow in this and the next chapter, we first find the resistive parameters  $R_L$ ,  $g_m$ ,  $r_\pi$ , and so on, that are required to make the network meet the midfrequency gain specification. Then we calculate the open-circuit resistances. From the specification on the high-frequency 0.707 point,  $\omega_h$ , we calculate  $\Sigma \tau_{jo}$  using Eq. 15.40. Now, using the relation

$$\sum \tau_{jo} = \sum R_{jo} C_j \quad (15.44)$$

we can select a transistor with appropriate values of  $C_\pi$  and  $C_\mu$ , keeping in mind such factors as the relative cost and availability of transistors to meet the requirements. This completes the first-cut design of the amplifier (other than biasing and the coupling and bypass capacitors).

Next we write the node equations for the mid- and high-frequency circuit model just derived, and calculate the poles, and  $\omega_h$  for the specified transfer function (voltage gain, for example). If the computed value of  $\omega_h$  is unacceptably far from the specified value, certain specific  $C_\pi$  and  $C_\mu$  values (hence transistors) can be changed. As we saw above, the open-circuit time constants provide an excellent guide as to which element should be changed. This design procedure is shown in flow-chart form in Fig. 15.6.